

OpenVoxel: Training-Free Grouping and Captioning Voxels for Open-Vocabulary 3D Scene Understanding

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Abstract

We propose **OpenVoxel**, a training-free algorithm for grouping and captioning sparse voxels for the open-vocabulary 3D scene understanding tasks. Given the sparse voxel rasterization (SVR) model obtained from multi-view images of a 3D scene, our OpenVoxel is able to produce meaningful groups that describe different objects in the scene. Also, by leveraging powerful Vision Language Models (VLMs) and Multi-modal Large Language Models (MLLMs), our OpenVoxel successfully build an informative scene map by captioning each group, enabling further 3D scene understanding tasks such as open-vocabulary segmentation (OVS) or referring expression segmentation (RES). Unlike previous methods, our method is training-free and does not introduce embeddings from a CLIP/BERT text encoder. Instead, we directly proceed with text-to-text search using MLLMs. Through extensive experiments, our method demonstrates superior performance compared to recent studies, particularly in complex referring expression segmentation (RES) tasks. The code will be open.

1. Introduction

With the recent advancement of the neural rendering algorithm [3, 5, 10, 11, 21, 27, 34, 38, 39, 43, 53, 56, 57, 60, 75], several primitives have been introduced to represent the 3D scenes. 3D Gaussian Splatting (3DGS) [24] and several extensions [6, 12, 69, 76] have recently emerged as a groundbreaking technique for novel view synthesis, offering an exceptional balance of visual fidelity and real-time performance. Concurrently, Sparse Voxel Rasterization (SVR) [29, 58] enables a sparse voxel representation to perform the novel-view synthesis task.

Recent studies [14, 15, 18–20, 22, 25, 28, 31, 33, 37, 46, 47, 49, 59, 68, 70, 71, 73, 74, 78] leverage the series of novel 3D primitives for open-vocabulary 3D understanding tasks, such as semantic segmentation, 3D open-vocabulary segmentation (OVS). LangSplat [49] is one of the pioneering papers that introduces 3D Gaussians for this purpose by optimizing 3DGS first and adopts Gaussian fea-

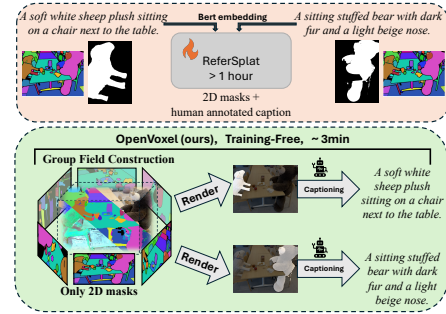


Figure 1. Comparison of lifting language into 3D representations between ReferSplat [15] and OpenVoxel. Note that ReferSplat additionally requires manual text annotations for training 3DGS to equipped with Bert embeddings, while ours is totally training-free and do not need any additional annotations.

tures that are trained to store language attributes derived from the 2D foundation model, CLIP [51]. The successive studies [19, 47, 71, 74, 78] try to improve the trainable Gaussian features by either introducing contrastive learning between different objects in the scene, or training an object-aware 3DGS model from scratch. However, these methods are limited by learning short words or tags, which hinders advanced 3D scene understanding with complex sentence queries such as the referring segmentation task, as mentioned in [15]. More recently, ReferSplat [15] proposes to cope with segmentation from complex queries, and focus on the 3D referring segmentation task (RES) by providing human-annotated observable objects’ masks and corresponding detailed natural language sentence descriptions for training. However, all the aforementioned approaches tend to leverage learned embeddings as language fields from visual-language aligned embeddings such as CLIP or DINO, limiting the capability of dealing with arbitrary textual queries [23]. Also, obtaining detailed human annotation for training requires labor-intensive pre-processing for each scene, and training additional language fields for 3D representation is also time-consuming (more than one hour for each scene).

To address these issues, we propose **OpenVoxel**, a novel training-free framework that bypasses the speed and ac-

curacy limitations of learning text embedding space. Instead of training 3D features to align with a fixed embedding manifold, our OpenVoxel directly populates the 3D scene with rich, human-readable text captions and proceeds with text-to-text retrieval for open-vocabulary 3D scene understanding, including open-vocabulary segmentation (OVS) and referring expression segmentation (RES). Given a Sparse Voxel Rasterization (SVR) model optimized from multi-view images of a 3D scene, we introduce a training-free algorithm of *Training-Free Sparse Voxel Grouping* to cluster all voxels into object-level groups by lifting and matching per-frame 2D segmentation maps obtained from SAM2 [52]. This aggregation is crucial for moving from a per-voxel representation to a semantically coherent, object-level understanding and ensuring view-consistency. To further bridge voxels with textual description, we also introduce the strategy of *Canonical Scene Map Construction* to locate language attributes on top of this sparse voxel representation via Vision Language Models (VLMs) and Multi-modal Large Language Models (MLLMs), obtaining a full description of the entire scene. Finally, to answer an open-vocabulary query (e.g., a simple word or a complex sentence), we utilize MLLMs to perform *Referring Query Inference* via direct text-to-text retrieval. The MLLM compares the user’s query against the constructed scene map, enabling a flexible and powerful retrieval process that is not constrained by embedding proximity. In summary, the contributions of our methods are listed below:

- *A training-free framework for open-vocabulary 3D scene understanding* : We introduce **OpenVoxel**, a novel pipeline that removes the need for learning 3D language embeddings and instead performs text-to-text reasoning using LLMs on top of the sparse voxel representation.
- *Group (object)-level captioning and scene map construction* : We propose an unsupervised method that clusters sparse voxels into instance-level 3D masks and generates rich, descriptive captions for each instance using an MLLM, forming an informative scene map.
- *Superior performance in both complex and simple language queries for 3D scene understanding task* : By combining instance captions with LLM-based reasoning, OpenVoxel outperforms previous studies on both open-vocabulary semantic segmentation (OVS) and referring expression segmentation (RES) tasks.

2. Related Works

3D scene representation with different primitives. Neural radiance fields (NeRF) [3, 5, 10, 11, 21, 27, 34, 38, 39, 43, 53, 56, 57, 60, 75] have established a powerful paradigm for representing scenes as continuous volumetric fields, enabling high-quality novel view synthesis but suffering from slow training and rendering. To address these limitations,

3D Gaussian Splatting (3DGS) [24] and following extensions [6, 12, 69, 76] represent scenes using anisotropic Gaussians that can be rendered in real time, significantly improving efficiency while preserving fidelity. More recent work on Sparse Voxel Rasterization (SVR) [58] introduces a discrete, sparse voxel representation that supports efficient rasterization-based rendering. These representations (i.e., NeRF, 3DGS, and SVR) collectively form the foundation for modern neural scene reconstruction and are increasingly adopted for downstream 3D understanding tasks.

Language-aligned 3D scene representation. A growing line of research [14, 15, 18–20, 22, 25, 28, 31, 33, 46, 47, 49, 59, 68, 71, 74, 78] aims to equip 3D scene representations with open-vocabulary or language-aware capabilities so that further downstream tasks [7, 8, 13, 16, 17, 32, 35, 40–42, 61, 63–66, 72] such as 3D scene editing or inpainting are applicable. LangSplat [49] introduces language-aligned features into 3D Gaussians by distilling CLIP embeddings and multi-scale SAM [26] masks from multi-view images, enabling open-vocabulary 3D segmentation. OpenGaussian [68] and Dr.Splat [22] further enhance this pipeline by improving the quality of language features in Gaussian primitives through codebooks. ReferSplat [15] tackles the challenging 3D referring expression segmentation (RES) task, aligning Gaussian features with sentence-level embeddings and predicting object masks conditioned on natural-language sentence queries. Despite promising results, these methods rely heavily on text-encoder embedding spaces and additional human annotation of “sentence-object mask” pairs on the training images, which limit their practicability on new scenes and ability to interpret nuanced or arbitrarily phrased queries.

In contrast, our method avoids the requirements of human-annotated description-mask pairs or embedding alignment entirely. Instead of training 3D features to match a fixed latent space, we generate rich captions for 3D instances and perform direct text-to-text retrieval using MLLMs, enabling more flexible reasoning over complex queries.

3. Preliminary

We start by reconstructing scenes from multi-view input images. To this end, we employ the recently emerged SVR [58], which reconstructs scenes as sparse voxel fields and achieves high-quality reconstruction and real-time rendering. The reconstructed voxels are also suitable for efficiently assigning attributes to certain 3D locations, which our algorithm exploits. Their rendering follows the classical volume rendering equation, which is also the one employed

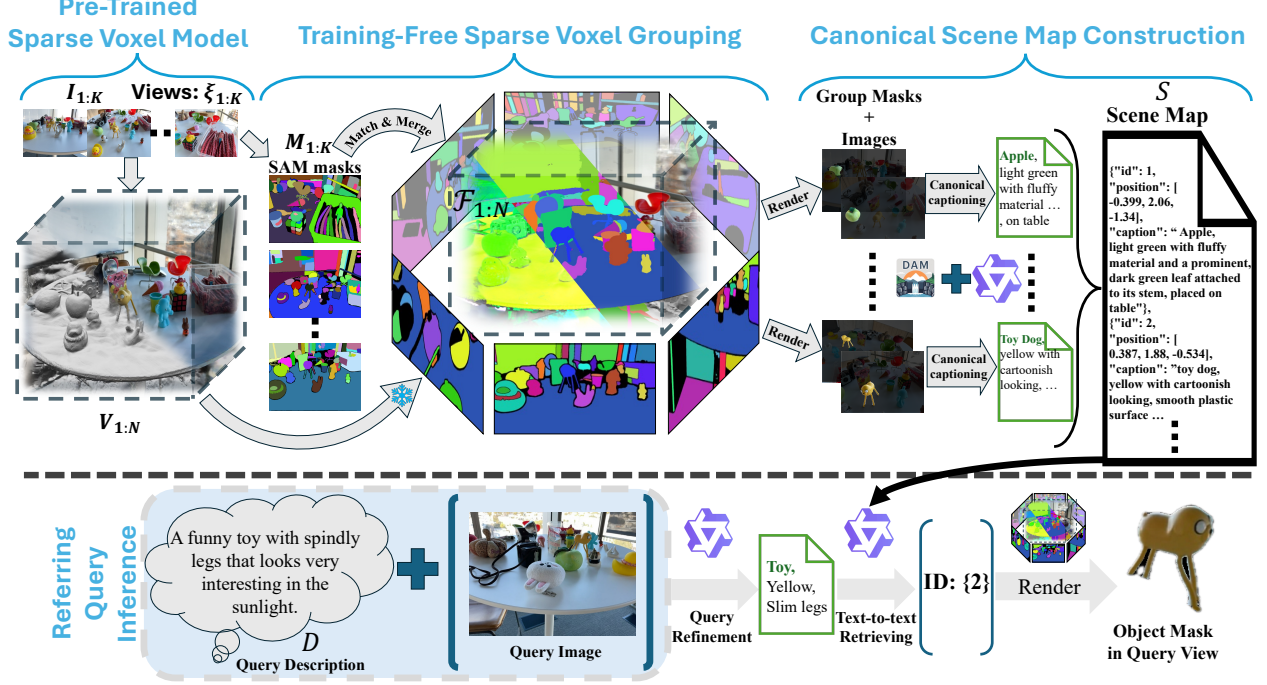


Figure 2. **Overview of OpenVoxel.** Taking a sparse voxel model $V_{1:N}$ pre-trained from multi-view images $I_{1:K}$ and their corresponding camera pose $\xi_{1:K}$, we aim to build a voxel group field $\mathcal{F}_{1:N}$ from segmentation masks $M_{1:K}$ obtained from SAM2. With $\mathcal{F}_{1:N}$ we render images and masks for all groups to obtain canonical captions and construct a Scene Map S recoding their position and captions. In the Referring Query Inference stage we take a query description and image to find the ideal target group that matches the description by query refinement and text-to-text retrieving from S , enabling complex segmentation tasks such as referring expression segmentation (RES).

by NeRF [39] and 3DGS [24]:

$$\mathbf{C}(\mathbf{r}) = \sum_{i < N} w_i(\mathbf{r}) \cdot c_i, \quad w_i(\mathbf{r}) = \alpha_i(\mathbf{r}) \cdot \prod_{j < i} (1 - \alpha_j(\mathbf{r})), \quad (1)$$

where $\mathbf{C}(\mathbf{r})$ is the rendering result for ray $\mathbf{r}$, i goes through an ordered list of N voxels in front of the ray, c_i is the voxel attribute (usually color) to render, $\alpha_i(\mathbf{r})$ is the opacity of the ray passing through the voxel, and finally $w_i(\mathbf{r})$ is the contributing weight from the voxel to the rendered ray, which is also used in our algorithm during grouping.

4. Methodology

We first define the notation and problem statement of our framework. As shown in Fig. 2, given a 3D scene reconstructed as N sparse voxels $\{V_i\}_{i=1}^N$ from a set of K multi-view images $\{I_i \in \mathbb{R}^{H \times W \times 3}\}_{i=1}^K$ with their camera poses $\{\xi_i \in \text{SE}(3)\}_{i=1}^K$, our goal is to assign descriptive language information to these voxels to construct a scene map S and enable open-vocabulary reasoning from a natural language description D , which can be a vocabulary as in 3D-OVS [33] or a referring expression as in 3D-RES [15]. We target gradient-descent-free and latent-dimension-free approaches to save space and time.

To this end, we present OpenVoxel, which consists of

three main components: (1) **Training-Free Sparse Voxel Grouping** that groups voxels into coherent 3D instances (Sec. 4.1), (2) **Canonical Scene Map Construction** that stores the detailed captions and 3D spatial information for all voxel groups (Sec. 4.2), and (3) **Referring Query Inference** to support querying and reasoning with natural language input from users (Sec. 4.3).

4.1. Training-Free Sparse Voxel Grouping

Drawing inspiration from deep hough voting [48] and spatial embedding [44], our key insight is that voxels belonging to the same object instance should cluster around a common 3D center position. With this motivation, we encode group membership of voxels as 3D spatial coordinates pointing to their instance centroids. These spatial embedding, termed as *group feature*, is progressively updated by readily available 2D segmentation masks from foundation models like SAM2 [52]. Our update is efficiently done in one training-free pass through all K views without the lengthy gradient descent. In contrast, existing methods like Gaussian Grouping [71] rely on gradient descent to update per-primitive high-dimensional features with sequential training processes.

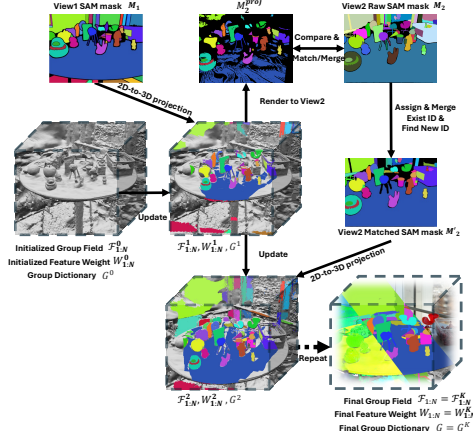


Figure 3. **Detail of the grouping process.** Taking the pre-trained voxel model $V_{1:K}$, we initialize the Group Field $\mathcal{F}_{1:N}^0$, Feature weight $W_{1:N}^0$ as empty tensors, and Group Dictionary G^0 as empty dictionary. Then start from ξ_1 , we project the SAM masks M_1 to 3D voxel and update $\mathcal{F}_{1:N}$, $W_{1:N}$, and G . By match and merge masks from the other views repeating this process, the final $\mathcal{F}_{1:N}$, $W_{1:N}$, and G is able to represent the group information of $V_{1:N}$.

Group Representations. As depicted in Fig. 3, given a reconstructed scene of N voxels, we extend voxel attributes by a group feature $\mathcal{F}_{1:N} \in \mathbb{R}^{N \times 3}$, which denotes the instance centroid the voxels belonging to, and a feature weights $W_{1:N} \in \mathbb{R}^N$ indicating the confidence about the current feature. We also use a Group Dictionary G to record the spatial centroid of each unique instance. The group features, weights, and dictionary are iteratively updated by the instance information from each frame.

Lifting 2D Segmentation Maps to 3D Group Field. We first introduce our method to lift a 2D instances mask into 3D voxels. Let M be the mask with m unique instances ID, and $\mathbf{f}^{\text{pts}} \in \mathbb{R}^{HW \times 3}$ be the rendered point map of the view, which indicates the expected ray-hit position in the world space. The centroid $\mathbf{f}_k^{\text{center}} \in \mathbb{R}^3$ of each instance k is then determined by performing masked average reduction on the point map:

$$\mathbf{f}_k^{\text{center}} = \frac{\sum_{j \in HW} \mathbb{1}(M_j = k) \cdot \mathbf{f}_j^{\text{pts}}}{\sum_{j \in HW} \mathbb{1}(M_j = k)}. \quad (2)$$

We then accumulate the instance centroids information into each voxel i by:

$$\begin{aligned} \mathcal{F}_i^{t+1} &= \mathcal{F}_i^t + \sum_{j \in HW} w_{ij} \mathbf{f}_{M_j}^{\text{center}}, \\ W_i^{t+1} &= W_i^t + \sum_{j \in HW} w_{ij}, \end{aligned} \quad (3)$$

where w_{ij} is the blending weight (Eq. (1)) the voxel i contributing to the pixel j . In practice, we customize the sparse

voxel renderer [58] to perform the update in one rendering pass. The update equation follows Dr.Splat [22], while we do not need the top-k sampling as our group feature is only 3-dimensional.

Start condition. At view ξ_1 , the voxel group feature $\mathcal{F}^1$ and weight W^1 are directly assigned by the information lifted from the view. The group dictionary is set to the m instance IDs and their corresponding centroids $\mathbf{f}^{\text{center}}$.

Progressively Matching and Merging Segmentation Masks. When we move on to view $t+1$, we need to first match the mask IDs in M^{t+1} with the existing voxel grouping $(\mathcal{F}^t, W^t, G^t)$. To this end, we render our group field into view $t+1$ as M_{t+1}^{proj} and match existing instances with those in M^{t+1} . The matched instance IDs are replaced with the existing IDs, while the remaining are added as new instances. Specifically, we first determine the instance ID of each voxel i in the current iteration by:

$$\text{ID}_i^t = \underset{j}{\text{argmin}} \left\| \frac{\mathcal{F}_i^t}{W_i^t} - G_j^t \right\|_2, \quad (4)$$

where the first term is the normalized voxel voting and the second term is the centroid of each instance. We then cast rays from the processing view to retrieve the instance ID of the hit voxel, which forms a instance mask M^{proj} reflecting the existing segmentation. Then, we conduct matching by finding the **highest** IoU mask in M^{t+1} for each instance in M^{proj} . We also prompt the SAM2 model again using M^{proj} and merge the instance masks that are largely overlapped to reduce noise and prevent the same instance from being incorrectly separated as two different IDs. As for the masks in M^{t+1} that are neither matched nor overlapped with the existing ones, we assign a new unique ID and add them to the group dictionary. Finally, we use the updated mask from M^{t+1} to update the voxel group feature into $(\mathcal{F}^{t+1}, W^{t+1})$ by Eq. (3). After processing all the K views, we use the ID^K from Eq. (4) as the final voxel instance grouping.

It is worth noting that, with our Group Field construction strategy, even if the voxels are assigned to an incorrect group in some view, the accumulated $\mathcal{F}_{1:N}$ and $W_{1:N}$ still force the voxel to “vote” the most confident group it belongs, reducing the possible miss-grouping.

4.2. Canonical Scene Map Construction

Once voxel groups are obtained, our next goal is to construct a canonical scene map S that records the position of each group and describes each group with a rich, human-readable caption, enabling open-vocabulary understanding of the entire scene (Fig. 2). Instead of training 3D features to align with language embeddings, we bypass this

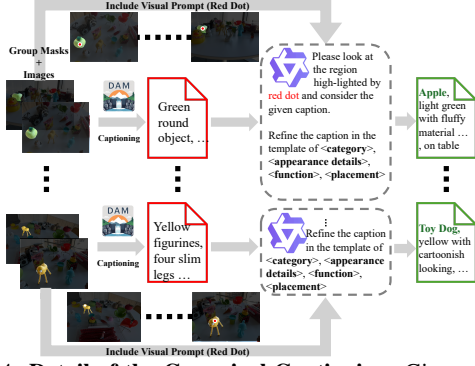


Figure 4. **Detail of the Canonical Captioning.** Given the group masks rendered of a specific group (taking the green apple as example) from our group field and their corresponding images, we leverage the Describe Anything Model (DAM) to first obtain a detailed caption. Then a Qwen3-VL model is conducted to canonicalize the caption into a fixed form, benefiting further usage.

bottleneck by leveraging Multimodal Large Language Models (MLLMs) to obtain detailed and accurate *canonical* captions for each group. We now detail the canonical captioning process.

As depicted in Fig. 2 and Fig. 4, taking the green apple as an example, we render the binary masks of the corresponding group in views $\xi_{1:K}$ and use them as mask inputs together with the original images $I_{1:K}$ for the Describe Anything Model (DAM) [30]. DAM produces a detailed description of the content inside each mask; however, its output is a free-form sentence and, in some cases, uses the generic word “object” as the subject (e.g., “A green round object, possibly an apple, ...”). To stabilize and standardize the captions across groups, we further use an MLLM (e.g., Qwen3-VL) to *refine* these free-form captions into a fixed-form caption while considering (i) the DAM sentence, and (ii) the original masked images.

Because general MLLMs are not specifically trained for mask-image captioning, we design the visual prompting strategy inspired by [67]: we darken the regions outside the mask and highlight the object with a small red dot to focus the model’s attention. We then instruct the MLLM to rewrite the caption into the canonical template $\langle \text{category noun} \rangle, \langle \text{appearance details} \rangle \langle \text{function/affordance or part-of} \rangle \langle \text{placement/relation} \rangle$. This converts each group’s description into a consistent, structured form. The resulting entry for a group in S stores its ID, 3D position (center), and the canonical caption. In practice, the canonicalization significantly reduces subject ambiguity (e.g., replacing “object” with “apple”) and yields stable, comparable descriptions across views.

4.3. Referring Query Inference

Given a user query D —from a single category token (e.g., “chair”) to a detailed referring expression (e.g., “what ob-

ject can be used for cutting paper?”)—and an optional query image of the target view, our goal is to identify the best-matching 3D instance in the scene map S , obtain its ID(s), and render the corresponding binary mask in the target view. Rather than mapping the query and all captions into a learned embedding space, we perform *direct retrieval* over S with a multimodal LLM (MLLM). This design avoids any additional training or calibration for our 3D setting, maintains robustness to free-form phrasing that includes attributes, affordances, or light reasoning, and yields interpretable evidence because the selection is made over the canonical captions stored in S .

Query Refinement. To make matching deterministic and stable, we first *canonicalize* the query. Following the same procedure as in our captioning stage, the MLLM rewrites D (with or without the query image) into our fixed template $\langle \text{category noun} \rangle, \langle \text{appearance details} \rangle \langle \text{function/affordance or part-of} \rangle \langle \text{placement/relation} \rangle$. For instance, the free-form description “A funny toy with spindly legs that looks very interesting in the sunlight” becomes “Toy, yellow, slim legs.” The refined query now matches the format of entries in S , so the retrieval process reduces to selecting the caption(s) in S that best satisfy the refined description and returning their associated ID (In practice, we input the entire S for the MLLM, so even if the query implies spatial relations (e.g., “left of the apple”), the MLLM can still possibly verify them using the stored group centers in S before finalizing the selection.

Text-to-text Retrieving of Target Group. Finally, given the selected ID(s), we render the referred object by rasterizing only the corresponding group(s) under the target view, producing a binary mask and the linked canonical caption from S . This inference procedure is fully training-free, applies uniformly to both OVS (category-only queries) and RES (attribute/affordance/relation queries), and empirically provides stable matches across scenes and phrasing styles while remaining simple to implement.

5. Experiments

5.1. Settings

Datasets. Following prior state-of-the-art works [15, 25, 49], we evaluate the open-vocabulary segmentation (OVS) and referring expression segmentation (RES) performance of our OpenVoxel primarily on subsets of the iPhone Polycam-captured LeRF dataset [25]. For the OVS task, we follow the evaluation protocols in [71, 78] using the *LeRF-Mask* subset, which contains three scenes (*ramen*, *figurines*, and *teatime*) with 6, 7, and 10 queryable objects, respectively. We further conduct experiments on the *LeRF-OVS* subset [49], with the same three scenes with 13, 17, and

Table 1. **Quantitative evaluation on the Ref-LeRF subset in terms of referring expression segmentation (RES) mIoU.** ReferSplat* denotes the best results we reproduced from the official implementation. Note that only ReferSplat requires ground truth annotation of description-mask pair for training. “Ours*” indicates all scenes share the same hyperparameters (from the “kitchen” scene), and “Ours” is our original results.

Method	Requires GT Annotation	ramen	figurines	teatime	kitchen	avg.
Grounded SAM [54]	-	14.1	16.0	16.9	16.2	15.8
LangSplat [49]	-	12.0	17.9	7.6	17.9	13.9
SPIn-NeRF [41]	-	7.3	9.7	11.7	10.3	9.8
GS-Grouping [71]	-	27.9	8.6	14.8	6.3	14.4
GOI [50]	-	27.1	16.5	22.9	15.7	20.5
ReferSplat* [15]	✓	31.0	20.0	25.4	21.4	24.5
ReferSplat [15]	✓	35.2	25.7	31.3	24.4	29.2
OpenVoxel (Ours*)	-	48.2	42.5	48.1	25.1	41.0
OpenVoxel (Ours)	-	52.5	43.5	48.4	25.1	42.4

Table 2. **Quantitative evaluation on the LeRF-OVS subset in terms of mIoU.**

Method	ramen	figurines	teatime	avg.
Feature-3DGS [77]	43.7	58.8	40.5	47.7
LEGaussians [55]	46.0	60.3	40.8	49.0
LangSplat [49]	51.2	65.1	44.7	53.7
GS-Grouping [71]	45.5	60.9	40.0	48.8
GOI [50]	52.6	63.7	44.5	53.6
3DVLGS [46]	61.4	73.5	58.1	64.3
ReferSplat [15]	55.1	67.5	50.1	57.6
CCL-LGS [59]	62.3	61.2	71.8	65.1
OpenVoxel (Ours)	62.5	60.7	75.4	66.2

12 annotated objects, respectively. For the RES task, we use the *Ref-LeRF* subset [15], which consists of four scenes (*ramen*, *figurines*, *teatime*, and *kitchen*) containing 13, 17, 12, and 11 objects, respectively, each paired with sentence-level natural language referring expressions.

Implementation details. We conduct our OpenVoxel with PyTorch [45] libraries, where our code base is built on top of SVR [58]. For the foundation models, we use SAM2 [52] to obtain the per-view segmentation map separately and also prompt SAM2 to conduct the mask merging operation during the Training-Free Sparse Voxel Grouping, as described in Sect. 4.1. On the other hand, the Describe Anything Model(DAM) [30] is used as our captioning model. As for the MLLM used to refine the obtained captions/query and conduct the target retrieving, we use Qwen3-VL-8B-Instruct [1, 2, 62]. Please refer to our supplementary material for more implementation details, including engineering tricks and the system prompt we use.

5.2. Quantitative Evaluations

Ref-LeRF. We first present the RES results on the Ref-LeRF subset in Table 1, as this task best demonstrates the advantages of our design. Since the official pre-trained models and data of the current state-of-the-art method,

ReferSplat [15], do not cover all four scenes, we reproduce its results following the original implementation and configurations, denoted as ReferSplat*. As shown in Table 1, despite OpenVoxel not requiring any description-mask annotations during grouping or scene map construction, it achieves a notable improvement of **13.2%** over the result reported in the original ReferSplat paper, and an even larger margin of **17.9%** compared to our reproduced version. During reproduction, we observe that ReferSplat’s strategy of learning sentence-level embeddings for 3D representations tends to overfit the seen descriptions, resulting in unstable evaluation performance. In contrast, our OpenVoxel consistently achieves higher accuracy, and the superior performance of our text-level retrieval mechanism supports its effectiveness and robustness compared to learned embedding approaches.

LeRF-OVS and LeRF-Mask. Tables 2 and 3 summarize the OVS results on the LeRF-OVS and LeRF-Mask subsets, respectively. For LeRF-OVS, we report mean Intersection-over-Union (mIoU) between predicted and ground-truth masks, following [15, 49]. For LeRF-Mask, we additionally evaluate mean Balanced IoU (mBIoU), following [71, 78]. As shown in Table 2, the OVS task involves relatively simpler queries, leading to higher overall performance across methods—even when evaluated on the same target objects as in Ref-LeRF. For LeRF-Mask, where the queries are fewer and less ambiguous, all current state-of-the-art methods (including ours) achieve over 70% mIoU. Although our design primarily targets the RES setting, by slightly change the prompt to OVS oriented version that emphasizes class name and appearance and verifies each group against the query, OpenVoxel still outperforms on both OVS benchmarks, demonstrating its flexibility and robustness across varying query complexities.

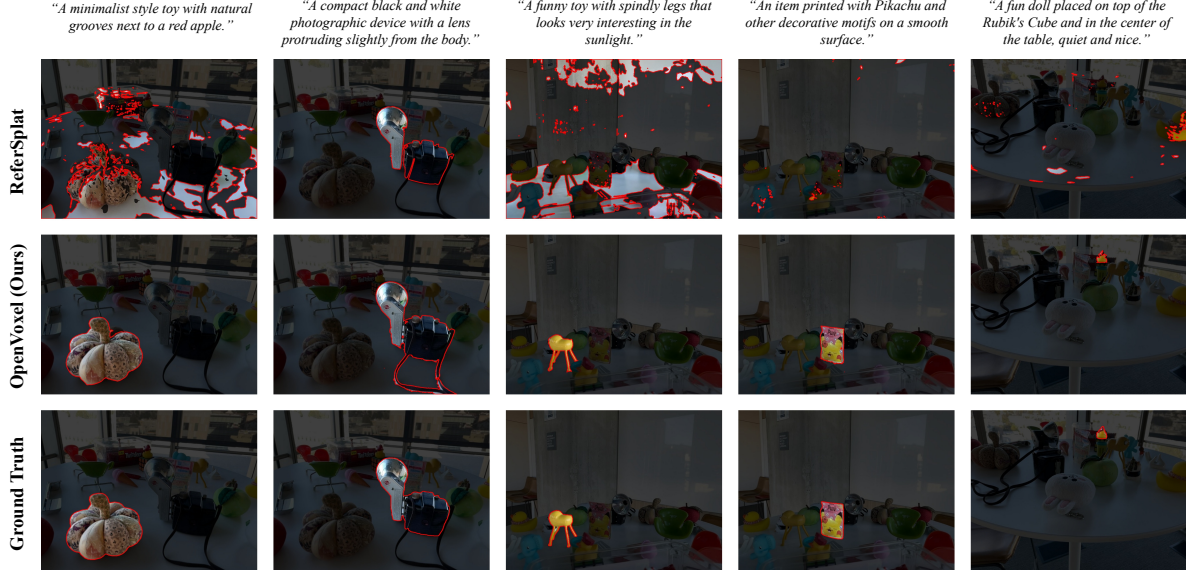


Figure 5. Qualitative results of RES task on the *Figurines* scene.

Table 3. Quantitative evaluation on the LeRF-mask subset in terms of IoU and BIoU. Note that BIoU indicates the boundary IoU of predicted mask and ground-truth mask. Best results are in bold text, and second best results are underlined.

Model	figurines		ramen		teatime		Avg.	
	mIoU	mBIoU	mIoU	mBIoU	mIoU	mBIoU	mIoU	mBIoU
DEVA [9]	46.2	45.1	56.8	51.1	54.3	52.2	52.4	49.5
LERF [25]	33.5	30.6	28.3	14.7	49.7	42.6	37.1	29.3
SA3D [4]	24.9	23.8	7.4	7.0	42.5	39.2	24.9	23.3
LangSplat [49]	52.8	50.5	50.4	44.7	69.5	65.6	61.2	56.1
GS Grouping [71]	69.7	67.9	77.0	68.7	71.7	66.1	72.8	67.6
Gaga [36]	<u>90.7</u>	<u>89.0</u>	64.1	61.6	69.3	66.0	78.5	74.2
ObjectGS [78]	88.2	85.2	88.0	<u>79.9</u>	<u>88.9</u>	88.6	<u>88.3</u>	<u>84.4</u>
OpenVoxel (Ours)	92.5	90.0	<u>87.7</u>	83.6	89.0	<u>86.9</u>	89.7	86.8

5.3. Qualitative Results

We show the qualitative results of “figurines” and “ramen” scene in Ref-LeRF subset in Fig. 5 and Fig. 6, respectively. In both scenes, ReferSplat often identifies only partial regions of the target object. For example, in Fig 5, when we input the query “A minimalist style toy with natural grooves next to a red apple”, ReferSplat focuses on the region near the apple (the apple is at the left of the image), without localizing the correct object. In contrast, OpenVoxel retrieves the toy pumpkin properly and render a smooth mask. As for the query “A drinking utensils with a smooth surface near a sake bottle” in Fig. 6, ReferSplat only focus on the term “drinking utensils” and incorrectly segments both the glass of water and the sake bottle, while our OpenVoxel locates the sake cup correctly. These results further support our

claim of using explicit captions to build a scene map and perform text-level retrieval, which benefits the RES tasks.

5.4. Further Analysis

Ablation Study. Table 4 shows the ablation studies of our OpenVoxel on the Ref-LeRF subset. Start from the baseline A where we do not merge overlapping small masks as described in Sect. 4.1 during the voxel grouping, and also do not canonicalize both the caption and the query. We can see that even with the explicit captioning and text-to-text retrieval, it is still hard to retrieve the correct target due to noisy small groups and misalignment of language format. After that, we include the mask merging strategy in model B, showing a slight improvement of 3.7% mIoU due to less noisy groups. Then, in model C, we introduce the canonical captioning for each group while constructing the scene

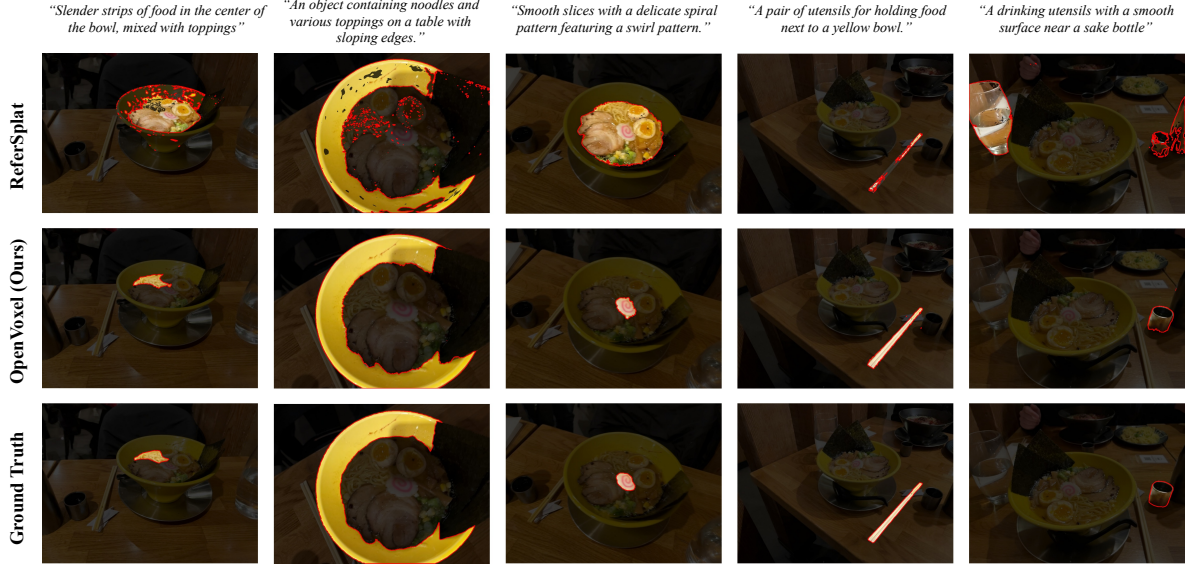


Figure 6. Qualitative results of RES task on the *Ramen* scene.

Table 4. Ablation study of our OpenVoxel on Ref-LeRF subset. We integrative evaluate the effectiveness of several components of our OpenVoxel.

Method	Mask Merging	Canonical Caption	Canonical Query	mIoU
A	-	-	-	24.3
B	✓	-	-	28.0
C	✓	✓	-	36.4
Ours	✓	✓	✓	42.4

map, showing a large improvement of 8.4% mIoU because of more specific captioning (i.e., standard template for each caption and reducing the word “object” as subject noun). Lastly, our full model canonicalizes the input text query into the same form as the canonical caption, introducing more precise alignment and less ambiguity, and showing the best performance. The above ablation study verifies the necessity of each of our designs.

Runtime Estimation. We show the comparison of run time between our OpenVoxel and current SOTAs on OVS and RES tasks in Table 5 on a single RTX 5090 GPU. The term “Requires Training” here indicates if a method needs training to obtain its feature/language field. Although ObjectGS [78] combines the training process of 3DGS and the per-Gaussian object feature (similar to Gaussian Grouping), we note that running a vanilla 3DGS on RTX 5090 only takes about 10 minutes, indicating ObjectGS needs an additional 30 minutes to obtain the object feature. On the other hand, although ReferSplat reports that it only takes 58 minutes to train on an A6000 GPU, in all our reproductions, we find that following the official implementation configuration takes at least 2 hours per scene to obtain the best results. In contrast, our OpenVoxel requires only about 3 minutes to complete the voxel grouping and canonical scene map con-

Table 5. Run time comparison with current SOTAs. For the term “Requires Training”, it is to indicate if a method needs gradient-based training to obtain the semantic or language field. All methods are reproduced on a single RTX 5090 GPU.

Method	Requires Training	Run Time
ReferSplat [15]	yes	>1 hr
ObjectGS [78]	yes	~ 40 min
OpenVoxel (Ours)	no	~ 3 min

struction, which is at least $10 \times$ faster than other SOTAs, and the inference time per query is less than 1 sec.

6. Conclusion

In this paper, we propose *OpenVoxel*, a training-free framework for grouping and captioning sparse voxel rasterization (SVR) models. In our *Training-Free Voxel Grouping* process, we curate a group field for a pre-trained SVR model by matching and merging per-view SAM2 masks. As for the *Canonical Scene Map Construction* we leverage VLMs and MLLMs to obtain a fix form of caption for each constructed group and then collect positions and captions of all groups as the scene map. And finally, we conduct a *Referring Query Inference* to retrieve target objects from our scene map with an input descriptive query. Through quantitative and qualitative experiments, we verify the efficiency and capability of OpenVoxel.

Acknowledgment This work is supported in part by the National Science and Technology Council via grants NSTC 114-2634-F-002-006 and NSTC 114-2221-E-002-056-MY2. We also thank the National Center for High-performance Computing (NCHC) for providing computational and storage resources.

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